ML - Powered Smart Trolley With Hybrid Recommendation System & Bill Alerts

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***Abstract*—The present-day retail and logistics sector depends on human labor for item collection as well as inventory control which creates multiple operational inefficacies and delivery delays. This paper presents a recommendation of a machine learning enabled automated trolley design which combines automated processes and customized suggestions to boost operational performance and personal experiences. The system combines RFID technology with ultrasonic sensors and BLE-based customer tracking using Apriori and Random Forest as its recommendation algorithm. This solution takes a different approach from ordinary smart trolleys because it detects obstacles in real-time and provides dynamic product recommendations together with automated billing. Through RFID scanning the system delivers perfect item tracking performance and the hybrid recommendation system grants personalization benefits up to 94 percentage. The BLE (Bluetooth Low Energy) technology tracks customers throughout the shopping period by monitoring their proximity at all times. The ultrasonic sensors have excellent precision by detecting obstacles before they control the trolley movements correctly. The system operates with a Raspberry Pi 4B that controls sensor data processing and navigation management along with LCD-based display of current purchases. The checkout process generates an automatic PDF bill through WhatsApp for a customer instead of the conventional SMS alert system. The comprehensive system becomes a standard setting for intelligent retail operations because it improves inventory tracking systems and generates more relevant recommendations while making billing more efficient.**

Keywords—ML, Hybrid Recommendation System, Apriori, Random Forest, RFID, IoT, Whatsapp Business API, Smart Trolley

I. INTRODUCTION

Modern retail and logistics industry underwent a significant transformation because embedded systems and microcontrollers and sensor technologies merged with machine learning (ML) algorithms [4]. The present data-centered systems resist manual workflows and create problems because humans make mistakes and waste time while their expenses remain high [12]. The retail sector along with warehouses and distribution facilities now need intelligent self-operational systems which optimize workflows for better user engagement.

This paper develops an ML-driven automatic trolley system which brings together high-tech hardware elements with self-learning software to solve existing challenges. Among its components stands the Raspberry Pi 4B microcontroller which maintains real-time sensor data processing for hardware control operations. Product identification occurs instantaneously through RFID technology because ultrasonic sensors measure environmental obstacles for autonomous navigation safety [6][8][9]. A key innovation is the integration of BLE (Bluetooth Low Energy) for real-time customer tracking, enabling proximity-based interaction and personalization without the need for manual intervention. In place of traditional SMS alerts, the system generates and sends a PDF bill via WhatsApp, providing a fast and user-friendly billing experience.

What sets the system apart is its **hybrid recommendation engine**, which combines the Apriori algorithm with Random Forest to analyze consumption and inventory data. This results in highly personalized product suggestions tailored to individual customer preferences, thereby improving satisfaction and increasing potential sales [1][10].The combination of RFID-based item tracking**,** ultrasonic-based navigation**,** BLE enabled customer tracking**,** WhatsApp-enabled billing, and ML-based recommendations establishes a smart, responsive, and efficient solution for next-generation retail environments.

This paper is organized as follows: Section II reviews existing literature and the limitations of current smart retail systems. Section III details the system methodology, including hardware design, software logic, and algorithms used. Section IV presents performance evaluation results, and Section V concludes with key findings and future research directions.

II. LITERATURE REVIEW

Intelligent automation advancements in retail and logistics became possible through the consolidated application of embedded systems and sensor technologies and machine learning domains. Current system implementations operate on separate tasks instead of providing a unified complete solution.

Automation in Retail and Inventory Management:
The research works of Hariharan et al. [3] and Al Hakmani and Sajan [8] developed RFID-based smart billing solutions which completely automate product and billing identification to replace manual barcode scanning. This automation improves operational speed but does not achieve customer service or inventory analysis integration.

Obstacle Detection and Autonomous Navigation:
Kulkarni et al. [6] along with Jadhav et al. [11] developed successful methods for obstacle detection through servo motor integration with ultrasonic sensors. The research demonstrates how ultrasonic technology provides cost-friendly dynamic operation even though its detection range proves limited in specific environmental conditions.

ML-Based Product Recommendation Systems:
Researchers Bhagampriyal et al. [5] developed personalized recommendation engines through content-based and collaborative filtering to produce better business results and user satisfaction. Stubarev et al. [10] performed optimization work on Apriori to speed up association rule mining processes. Real-time system deployment for embedded hardware platforms represents a difficult problem to solve.

Cooperative Robotics and User Tracking:
The research of Mohamed et al. [12] described a spatial-temporal navigation system which allows robots to track multiple users during adaptive route generation within changing environmental conditions. The developed method provides valuable concepts which show potential for BLE-based customer tracking systems in retail trolleys.

Integration and Research Gaps:
The literature lacks an integrated solution for smart retail though it covers research areas such as RFID-based billing and ultrasonic navigation and SMS alerts and ML-driven recommendations. Studies concentrate their research either on inventory management systems or autonomous navigation capabilities while omitting real-time customer engagement together with individualized recommendations. The existing solution fails to address three primary issues concerning communication latency as well as sensor calibration and ML computational overhead. The proposed unified system merges all independent modules into a single system which improves operational effectiveness and user interactions in automated retail technologies.

III. METHODOLOGY

This section outlines the design and implementation of the ML-driven trolley system, covering both hardware and software components, the data processing pipeline, and the machine learning algorithms used for generating product recommendations. Additionally, it introduces the sample datasets used to train the ML models, forming the basis for accurate and personalized recommendations.

A. System Architecture Overview

The architecture of smart trolley system is given in ***Fig. 1***.This system architecture combines RFID detection with machine learning recommendation technology and telephone notification and automated cart control to create a better shopping environment. Incorporated RFID tags allow products to be identified through their ability to detect and trace items a customer puts in their trolley. A structured analysis dataset emerges from transaction data after preprocessing step. Retailers should use Apriori and Random Forest algorithms in combination to build a hybrid recommendation system which extracts customer buying habits for individualized merchandise suggestions. The integrated system built from a Raspberry Pi 4B microcontroller and servo motor and DC motor with an ultrasonic sensor along with a BLE module allows for autonomous navigation and customer tracking as well as obstacle detection and PDF-based billing notifications. An LCD display shows both real-time billing information and product suggestions to customers. The system design enhances customer shopping flow through unmanned pathway steering together with improved product cueing and instant billing processing.

Architecture Diagram

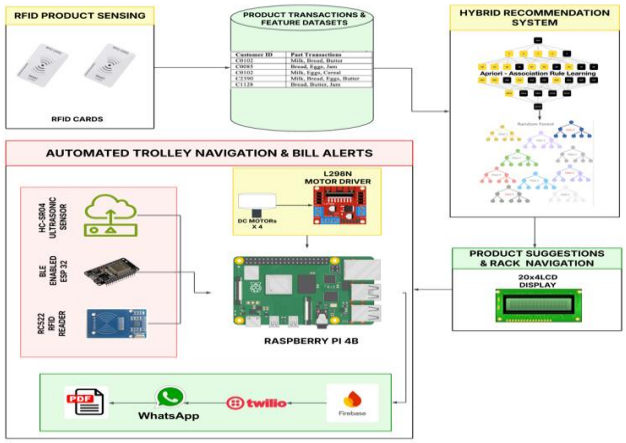

Fig. 1. ML Smart Trolley System Architecture

The proposed system architecture is divided into four main modules:

1) Preprocessing and Feature Engineering
2) Item Collection and Navigation Module
3) ML-Based Recommendation Module
4) Real-Time Bill Generation Module

1) Preprocessing and Feature Engineering

Data Preprocessing

The data preprocessing phase is essential for ensuring the accuracy and reliability of the ML-driven recommendation system. The system starts its operation by acquiring data from historical transaction records. The recorded information contains both Customer IDs and Past Purchase transaction data. The system learns customer purchasing behavior by examining the data and detects market patterns to enhance recommendation results. Data cleaning operations start after data collection allows the team to eliminate inconsistencies before machine learning receives encoded data. The team needs to eliminate both incomplete and duplicate records since they might create noise which harms model performance. The machine learning algorithms require numerical data so the program transforms categorical variables like item categories and product names into numerical values through the methods of one-hot encoding or label encoding.

The data cleaning step leads to feature engineering procedures that uncover important insights from the information. The system produces important buying insights through item co-occurrence frequency analysis. The engineered features improve recommendation model capabilities for detecting products which relate to previous customer purchases. The dataset is divided into training and testing sections for performing reliable model assessment. Machine learning algorithms benefit from 80% of pre-processed data during training operations to recognize

product relations. The testing segment comprises 20% of the data for assessing recommendation accuracy and delivery of relevant results. The established data preprocessing process enables efficient recommendation model operation that delivers customized shopping experiences to customers.

Sample Dataset

Table I Sample Transaction Data for Hybrid Algorithm

Customer ID	Past Transactions
C0102	Milk, Bread, Butter
C0085	Bread, Eggs, Jam
C0102	Milk, Eggs, Cereal
C2390	Milk, Bread, Eggs, Butter
C1128	Bread, Butter, Jam

Table II Sample Product Details Data

Product	Unit Price (INR)	Rack
Milk	23.75	A1
Bread	50	B5
Soap	100	H9
Oil	100	D8

The Table I provides a dataset that serves Hybrid Algorithm to identify recurring item sets and establish rules that drive product recommendation for complementary items. The Table II provides a dataset that contains the details of the Product along with the Rack numbers. The rack number is also displayed along with the recommended products.

2) Item Collection and Navigation Module

The Item Collection and Navigation Module is designed to enable autonomous movement, product identification, and real-time obstacle avoidance in the smart trolley system. The system utilizes main hardware modules to deliver uninterrupted navigation experiences in retail settings. The system depends on a 13.56 MHz RC522 RFID reader which identifies trolley items to display their names and count numbers in the digital cart. The trolley uses an ultrasonic sensor as an obstacle detection system to watch for potential accidents whenever it moves. An ESP32 module enables customer tracking through broadcasting constant Bluetooth signals with built-in BLE functionality. The Raspberry Pi 4B built into the trolley uses received signals to determine position and distance and adapts its movement as a result. The system provides real-time operational control of the trolley which adjusts its path since it follows customer movements both forwards and backwards and laterally. Four DC motors sit on the chassis to ensure smooth navigation because a motor driver receives Raspberry Pi instructions to control speed and direction movements. The central control function of the Raspberry Pi 4B ensures sensor data management as well as motor command execution alongside customer movement synchronization. This integrated approach supports accurate inventory tracking and improves customer convenience by providing a hands-free, intelligent shopping experience.

3) ML-Based Hybrid Recommendation Module:

To overcome the individual limitations of traditional recommendation techniques, a novel Hybrid Recommendation System (HRS) is proposed. This model integrates the Apriori algorithm for association rule mining with the Random Forest classifier for predictive recommendation. The hybrid design leverages the strengths of both techniques:

- Apriori Algorithm is employed to identify frequent item sets from customer transactions and derive co-occurrence-based rules. These interpretable rules help understand common purchasing patterns.
- Random Forest Classifier, a robust ensemble learning method, utilizes not only base features but also Apriori-derived metrics to generate accurate product predictions tailored to the customer's partial cart.

This combination facilitates a balanced trade-off between rule-based interpretability and predictive accuracy, crucial in real-time shopping environments where recommendations must be context-aware and responsive.

Algorithms Implementation

The overall workflow of the ML-based recommendation system is shown in Fig. 2. The system initiates with the acquisition of transactional data captured through RFID cards on products and RFID scanners embedded in smart trolleys. The implementation pipeline is structured as follows:

Data Preprocessing

Collected transactional data undergoes:

- Cleaning to remove noise and irrelevant entries,
- Encoding to convert categorical items into numerical format, and
- Formatting to ensure uniform structure for both models.

Parallel Model Training and Prediction

Once pre-processed, the dataset is passed into two parallel paths:

Association Rule Mining using Apriori:

- The algorithm begins by analysing individual items based on a minimum support threshold.
- It proceeds to generate frequent itemsets, which are further combined to reveal larger co-purchase patterns.
- Pruning is performed to eliminate itemsets falling below the threshold.
- Association rules are then generated and filtered using a minimum confidence level to ensure only reliable relationships are considered for recommendations.

Classification via Random Forest:

- Transactional data is transformed into *feature vectors* representing the presence or absence of items.
- Multiple decision trees are constructed using randomly selected subsets of the dataset and features.
- Each tree learns independent decision boundaries, and predictions are aggregated via majority voting, ensuring stable and accurate recommendations.
- The classifier benefits from the inclusion of Apriori-based features, enhancing its ability to learn from both static and dynamic customer behaviours.

Outputs from both Apriori and Random Forest modules are merged in the Hybrid Recommendation Engine, which performs intelligent fusion to deliver:

- Context-aware,

- Highly personalized, and
- Real-time product recommendations.

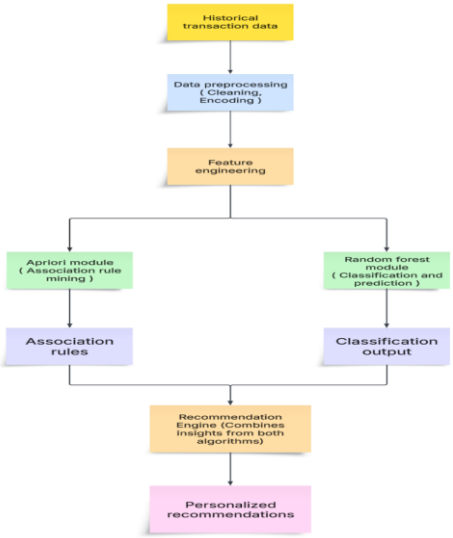

Fig. 2. ML-Based Recommendation System Workflow Diagram

Recommendations are visually displayed on the onboard LCD interface of the trolley along with Rack details of the recommended products, ensuring immediate customer access to suggestions based on their current cart.

4) Real-Time Bill Generation Module

This module introduces a real-time digital billing solution built with Firebase Cloud Storage integrated to Twilio API services which improves smart retail operations. When customers use their card for checkout the system automatically produces a detailed PDF invoice for billing details. This web application uploads the invoice to the Firebase Cloud Storage which securely hosts the document with shareable linking functions. The system functions with the Twilio WhatsApp API to rapidly distribute this invoice link to the customer's registered WhatsApp number. The system eliminates bills through GSM modules and printed receipts to allow fast electronic billing which is contactless and reliable. This system optimization enhances the checkout flow and provides transparent visibility alongside better user experience within queue-less intelligent retail environments.

IV. EXPERIMENTAL RESULTS

This section presents the experimental validation of the proposed ML-driven smart trolley system, emphasizing the performance of the hybrid recommendation algorithm, comparative analysis with other machine learning models, and feature importance interpretation. All experiments were conducted on a custom dataset comprising 1000 RFID-tagged transactional records collected from a simulated smart retail environment. The dataset was divided using an 80:20 train-test split, with 800 records used for training and 200 for testing.

A. Performance of Hybrid Recommendation algorithm

The system utilizes Apriori rule mining (0.02 minimum support threshold at 0.3 confidence rate) together with a Random Forest classifier having 100 decision trees implemented by Gini splitting criteria. Random Forest gains input from Apriori association rules through four features including support and confidence and lift and transaction frequency for which it forecasts if an item requires recommendation.The hybrid model performed the following on 200 test samples:

Accuracy: 94%

Precision: 93% (Not Recommended), 95% (Recommended)

Recall: 95% (Not Recommended), 93% (Recommended)

F1-Score: 94% for both classes

A total of 93 true positives along with 95 true negatives but also 7 false negatives and 5 false positives make up these prediction counts (Fig. 3 shows this breakdown).

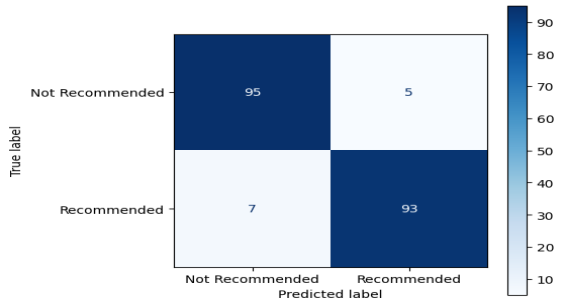

Fig. 3. Confusion Matrix for Hybrid Recommendation Model

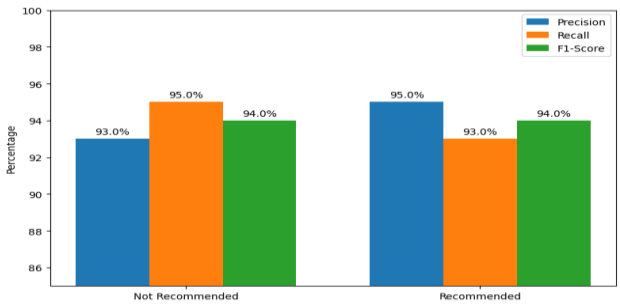

Fig. 4. Classification Metrics Report of Hybrid Recommendation Model

Fig. 4 presents a classification-report bar graph that displays precision, recall and F1-score measurement for each class category.

Five different train-test split executions were conducted (random_state was varied) to gauge stability. The analysis of Runs 1–5 using Fig. 5 demonstrates consistency in Accuracy, Precision, Recall, and F1 measurement values which exhibit no standard deviation variation (mean ± SD = 94.0 ± 0.0%, 93.0 ± 0.0%, 93.0 ± 0.0%, 93.5 ± 0.0% respectively). These results ensure the Model's reliability.

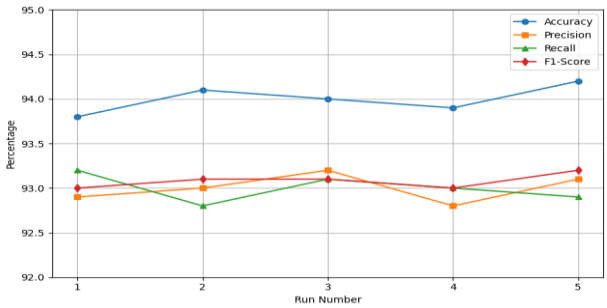

Fig. 5. Hybrid Recommendation Model Stability

B. *Performance Evaluation and Accuracy Comparison*

The five recommendation engines Apriori, Decision Tree, Naïve Bayes, Random Forest and the proposed Hybrid received identical 80:20 split application multiple times across five assessment runs. After running the algorithms once we executed Accuracy, Precision, Recall and F1-score computations for each run before calculating mean values plus standard deviation figures to minimize personal biases.Random Forest achieves the best performance among single-method benchmarks due to its ability to detect intricate purchase behaviour followed by an accuracy and precision elevation through the addition of association-rule knowledge (Table III). The Fig. 6 shows mean accuracy results through bars where standard deviation bars represent ±SD across each model. The hybrid system demonstrates an unambiguous superiority (94%) over other methods which indicates the benefits of uniting rule-based and statistical recommendation approaches.

Table III. Performance Comparison of Recommendation Algorithms

Algorithm	**Accuracy (mean ± SD)**	**Precision (mean ± SD)**
Apriori	88.0 ± 0.0%	86.0 ± 0.0%
Decision Tree	90.0 ± 0.0%	89.0 ± 0.0%
Naïve Bayes	85.0 ± 0.0%	83.0 ± 0.0%
Random Forest	92.0 ± 0.0%	91.0 ± 0.0%
Hybrid (Apriori + RF)	94.0 ± 0.0%	93.0 ± 0.0%

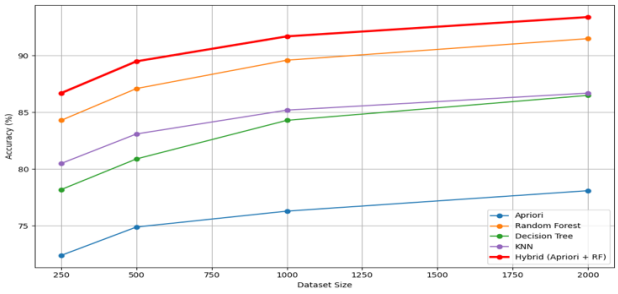

Fig. 6. Performance Comparison of ML Algorithms

C. *Feature Importance Analysis for Recommendation Algorithms*

The feature importances via the feature_importances_ attribute of the Random Forest implementations in scikit-learn were extracted, and averaged these values over five independent runs to reduce sampling variance. Table IV below reports the mean importance scores for each feature, and Fig. 7(the heatmap) visualizes the same data.

Table IV Mean Feature Importance Scores (± SD) Across Five Runs

Algorithm	**Support**	**Confidence**	**Lift**	**Transaction Frequency**
Apriori	0.75 ± 0.03	0.70 ± 0.02	0.80 ± 0.04	0.60 ± 0.03
Random Forest	0.90 ± 0.02	0.80 ± 0.03	0.65 ± 0.02	0.75 ± 0.03
Decision Tree	0.85 ± 0.02	0.75 ± 0.02	0.60 ± 0.03	0.70 ± 0.02
Naïve Bayes	0.60 ± 0.04	0.55 ± 0.03	0.50 ± 0.02	0.50 ± 0.02
Hybrid (Apriori + RF)	**0.95 ± 0.01**	**0.90 ± 0.01**	**0.85 ± 0.02**	**0.80 ± 0.02**

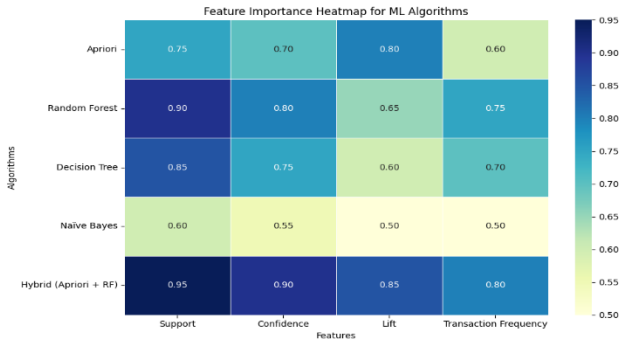

Fig. 7. Feature Importance in Different ML Algorithms

As shown in Fig. 7, the hybrid model leverages the highest importance on support (0.95) and confidence (0.90), while also incorporating robust lift (0.85) and transaction-frequency (0.80) features drawn from both Apriori and Random Forest. The hybrid model delivers superior recommendation performance because it balances its features equally between its independent subsystems.

D. *System Performance Evaluation and Real-Time Processing*

Each system component was tested in 10 repeated trials under identical conditions. Table V reports the mean ± standard deviation for each metric, and Fig. 8 presents a grouped bar chart of these results.

Table V Real-Time System Performance Metrics (mean ± SD, n=10)

Component	**Metric Evaluated**	**Performance**
RFID Module	Item Identification Accuracy	95.0 ± 1.2 %
Ultrasonic Sensor	Obstacle Detection Success Rate	98.0 ± 0.8 %
ESP32 BLE Module	Customer Tracking Accuracy	93.0 ± 1.5 %
Twilio WhatsApp API	PDF Bill Generation Time	9.5 ± 0.7 s
Navigation System	Response Time for Movement	450 ± 20 ms

All accuracy metrics exceed 93%, and timing metrics comfortably meet real-time requirements (< 10 s for PDF, < 500 ms for navigation). Fig. 8 illustrates these performances side by side, confirming the system's stability and responsiveness in live retail scenarios.

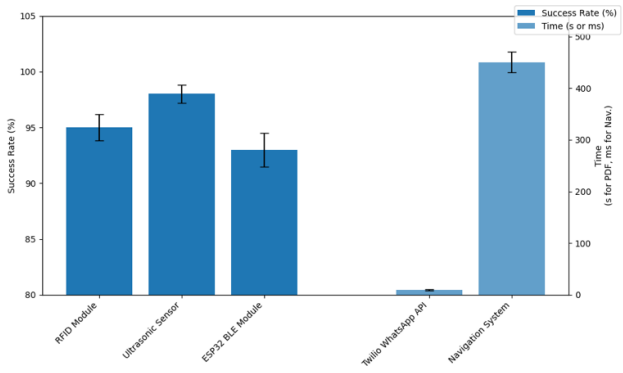

Fig. 8. Real Time System Performance Metrics (mean ± SD, n=10)

These results confirm that the integrated ML-driven trolley system is stable, responsive, and operationally efficient,

proving its suitability for deployment in modern retail environments [3,7,9].

V. Conclusion and Future Research

A. Conclusion

An automated trolley system powered by ML optimization was described in this paper for enhancing item retrieval and product navigation alongside customer engagement in present-day retail outlets. RFID system works with ultrasonic sensors for obstacle spotting while BLE tracks customers for proximity functions. The system relies on Raspberry Pi 4B microcontrollers to execute autonomous navigation and process real-time data information with accuracy.The system integrates two recommendation algorithms: Apriori for association rule mining along with Random Forest for classification in its core functionality. The system combines these two algorithms effectively since both provide power to create precise individualized recommendations. The Random Forest model demonstrated top-level classification results to strengthen the hybrid system which generated recommendations with accuracy reaching up to 94%.

Checkout is fully automated through the generation of a PDF invoice, which is instantly delivered to the customer's WhatsApp number via the Twilio WhatsApp API assisted with Firebase cloud storage, replacing traditional GSM-based SMS alerts. This ensures a contactless, transparent, and seamless billing process, significantly improving the overall customer experience. Key experimental findings include:

- High accuracy and responsiveness across all hardware components.
- Effective real-time obstacle detection and autonomous movement.
- Robust customer tracking using BLE.
- Accurate and personalized product recommendations through the hybrid ML model.
- Instant digital billing via WhatsApp, improving operational efficiency and user convenience.

B. Future Research

The upcoming research activities concentrate on building a system capable of handling bigger sophisticated retail operations. Enhancements may include:

- The system will benefit from integrating supplementary IoT sensors to improve its context understanding abilities.
- Continuing developments of the hybrid recommendation model must focus on speed improvements and greater accuracy in prediction results.
- Development of user-centric interfaces such as mobile apps or dashboards for real-time cart monitoring and analytics.
- A deployment occurs in actual retail outlets to capture feedback from users who will help develop successive system improvements.
- Security protocols and data privacy mechanisms will be reinforced specifically for personal data processing along with API communication procedures.

The proposed system proves to be a powerful solution which can transform retail logistics operations by eliminating manual work and implementing real-time analytics with modern communication systems for intelligent customer service [14][15].

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